Efficiency

Question 1

```
set.seed(43)

rbeta(40, shape1 = 2, shape2 = 0.7)
```

```
 [1] 0.7239814 0.4825235 0.8929028 0.6023222 0.8739952 0.7134292 0.3663387
 [8] 0.2230526 0.5597774 0.9761917 0.9137023 0.9987036 0.9912153 0.8194507
[15] 0.3469107 0.6652637 0.9407395 0.5539475 0.7362972 0.9167468 0.5935360
[22] 0.7017824 0.3652604 0.9962011 0.8909513 0.6857335 0.7225687 0.9922536
[29] 0.9585488 0.8544784 0.5243689 0.2472969 0.9103013 0.2998932 0.9064048
[36] 0.8744211 0.6209194 0.4749027 0.7553726 0.8484121
```

```
samples <- replicate(1000, rbeta(40, shape1 = 2, shape2 = 0.7))
```

Question 2

```
add <- function(a, b){
  print(a + b)
}

add(2, 2)
```

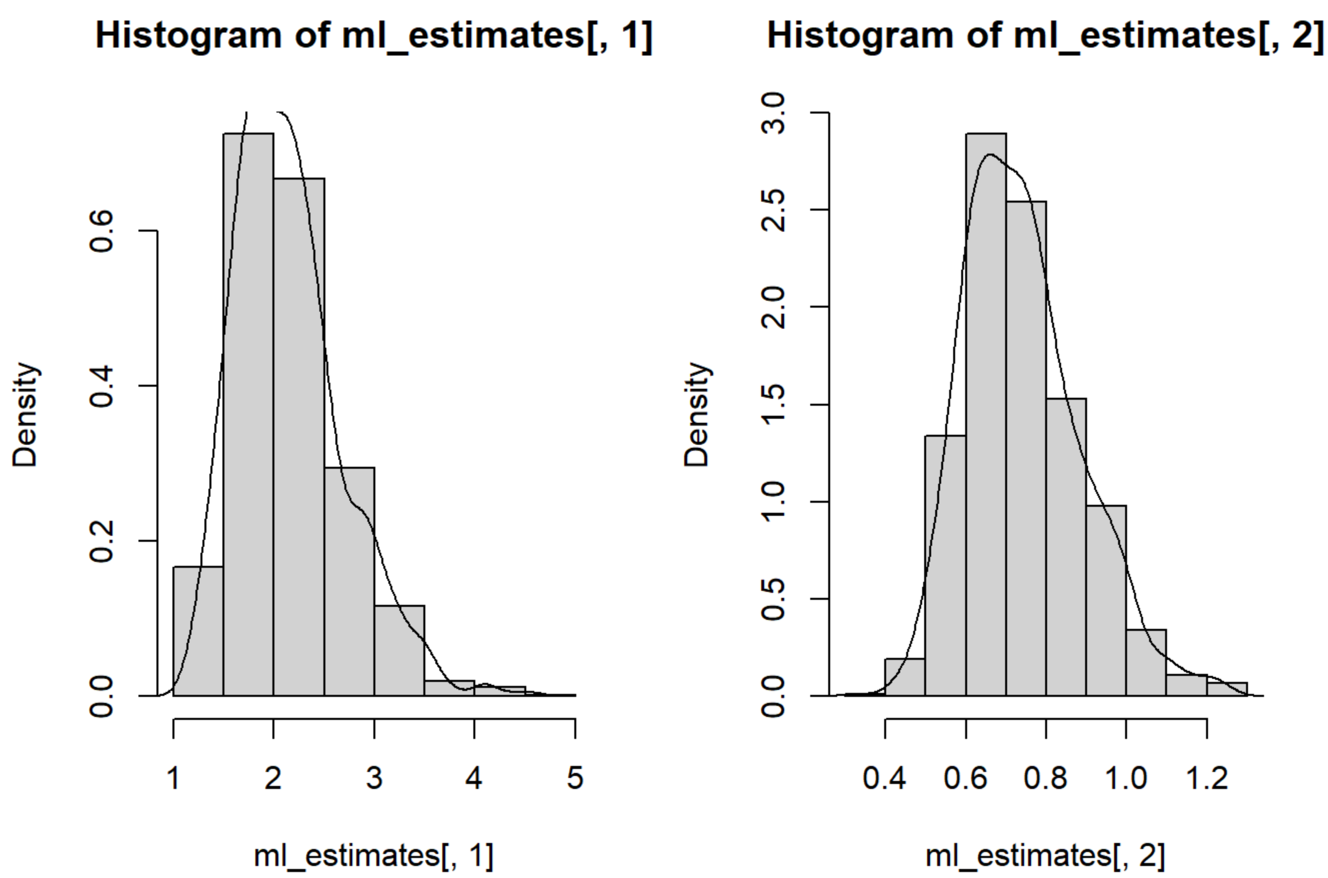
```
[1] 4
```

```
## installing
#install.packages("fitdistrplus")

## load package
library(fitdistrplus)
```

```
Warning: package 'fitdistrplus' was built under R version 4.5.3

Loading required package: MASS

Loading required package: survival
```

```
estbeta.ml<-function(y)
{
  coefficients(fitdist(y,distr="beta"))
}

ml_estimates <- apply(samples, 2, estbeta.ml)
```

```r
ml_estimates <- t(apply(samples, 2, estbeta.ml))

par(mfrow = c(1,2))

hist(ml_estimates[,1], prob = TRUE)

lines(density(ml_estimates[,1]))

hist(ml_estimates[,2], prob = TRUE)

lines(density(ml_estimates[,2]))
```



Question 4

```r
# alpha
sd(ml_estimates[,1])
```

```
[1] 0.5398479
```

```r
# beta
sd(ml_estimates[,2])
```

```
[1] 0.1460461
```

Question 5

```r
estbeta.mm <- function(y){
  m <- mean(y)
  v <- var(y)
  alpha_mm <- m * ((m * (1 - m)/v)-1)
  beta_mm <- (1 - m)*(((m * (1 - m))/v)-1)
  c(shape1 = alpha_mm, shape2 = beta_mm)
}

mm_estimates <- apply(samples, 2, estbeta.mm)

mm_estimates <- t(apply(samples, 2, estbeta.mm))


se_mm_alpha <- sd(mm_estimates[,1])
se_mm_beta <- sd(mm_estimates[,2])

se_mm_alpha
```

```
[1] 0.5895088
```

```r
se_mm_beta
```

```
[1] 0.1765054
```

The TcCB Data

Question 1

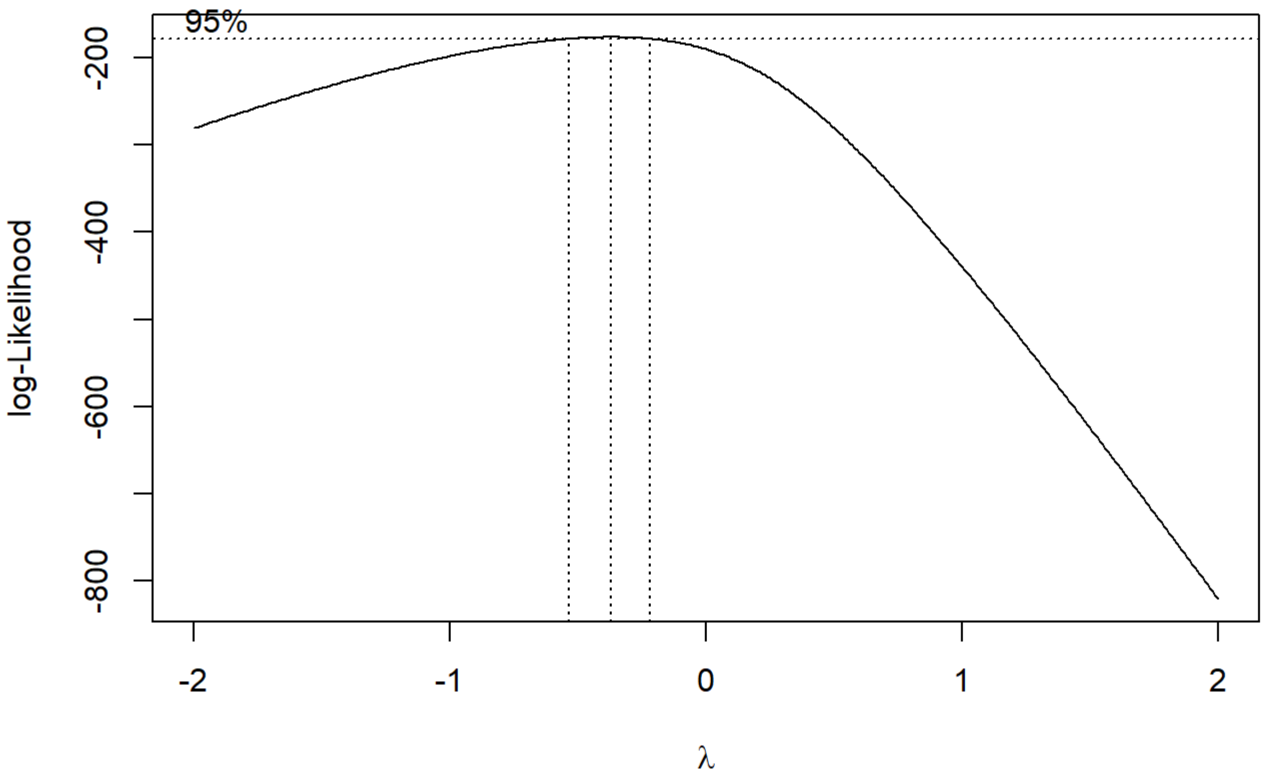
```r
EPA.94b.tccb.df <- read.csv("EPA.94b.tccb.df.txt")
```

Question 2

```r
cleanup <- subset(EPA.94b.tccb.df, Area == "Cleanup")

cleanup <- cleanup$TcCB
```

Question 3

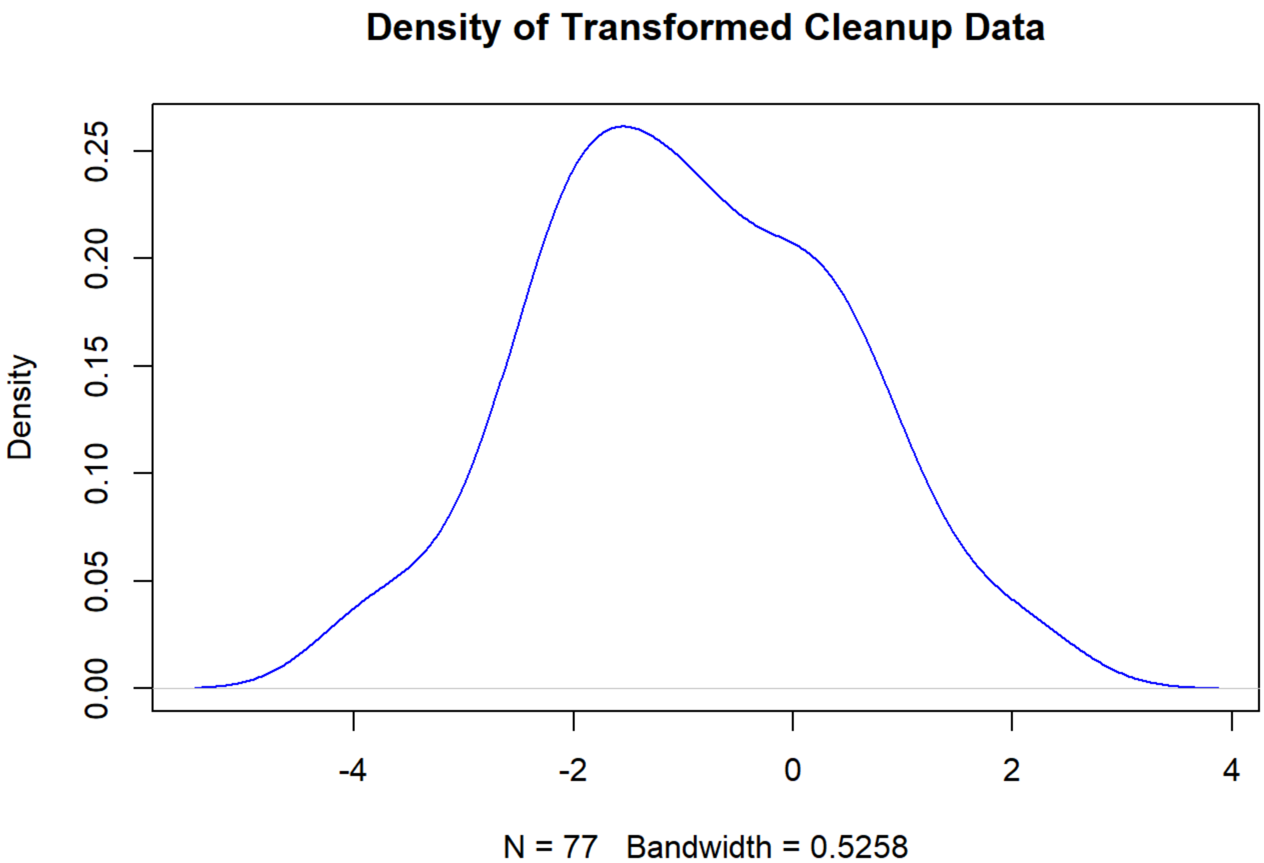
```r
bc <- boxcox(cleanup ~ 1, lambda = seq(-2,2,by = 0.01))
```



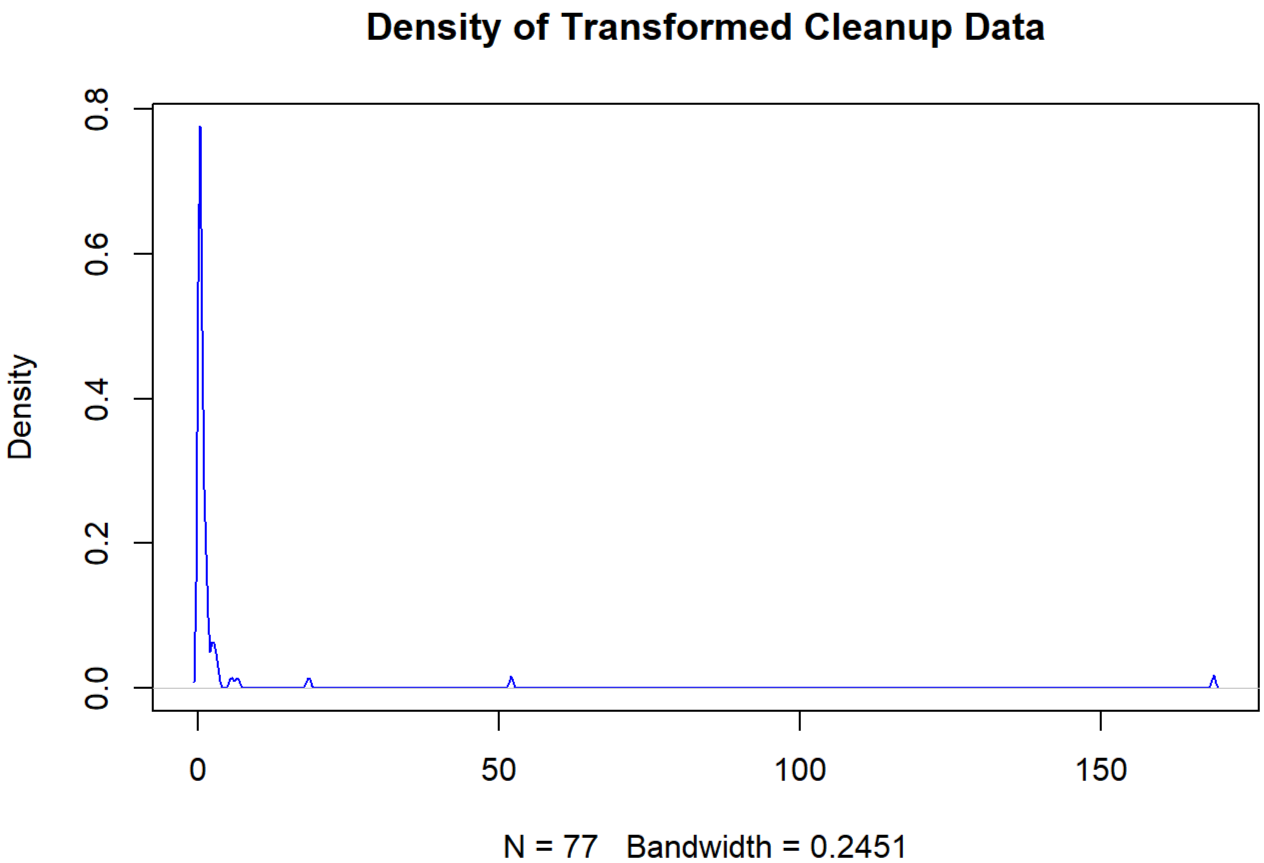

```r
which.max(bc$y)
```

```
[1] 164
```

```r
lambda_opt <- bc$x[which.max(bc$y)]

cleanup_transformed <- (cleanup^lambda_opt - 1)/lambda_opt


# plot a density for the above histogram
plot(density(cleanup_transformed),
     col = "blue",
     main = "Density of Transformed Cleanup Data")
```




```r
plot(density(cleanup),
     col = "blue",
     main = "Density of Transformed Cleanup Data")
```



Question 4

```
library(fitdistrplus)

fit <- fitdist(cleanup_transformed, "norm")
fit
```

```
Fitting of the distribution ' norm ' by maximum likelihood
Parameters:
       estimate Std. Error
mean -0.9499296  0.1576865
sd    1.3836933  0.1115009
```

```
mean_estimate <- -0.9499
sd_estimate <- 1.3837
```

Question 5

```
q99_transformed <- qnorm(0.99,
                         mean = mean_estimate,
```

```
                                  sd = sd_estimate)
q99_transformed
```

```
[1] 2.269068
```

Question 6

```
a <- exp(q99_transformed)
a
```

```
[1] 9.670379
```

Question 7

```
p = 0.01

pbinom(0, size = 3, prob = 0.01, lower.tail = FALSE)
```

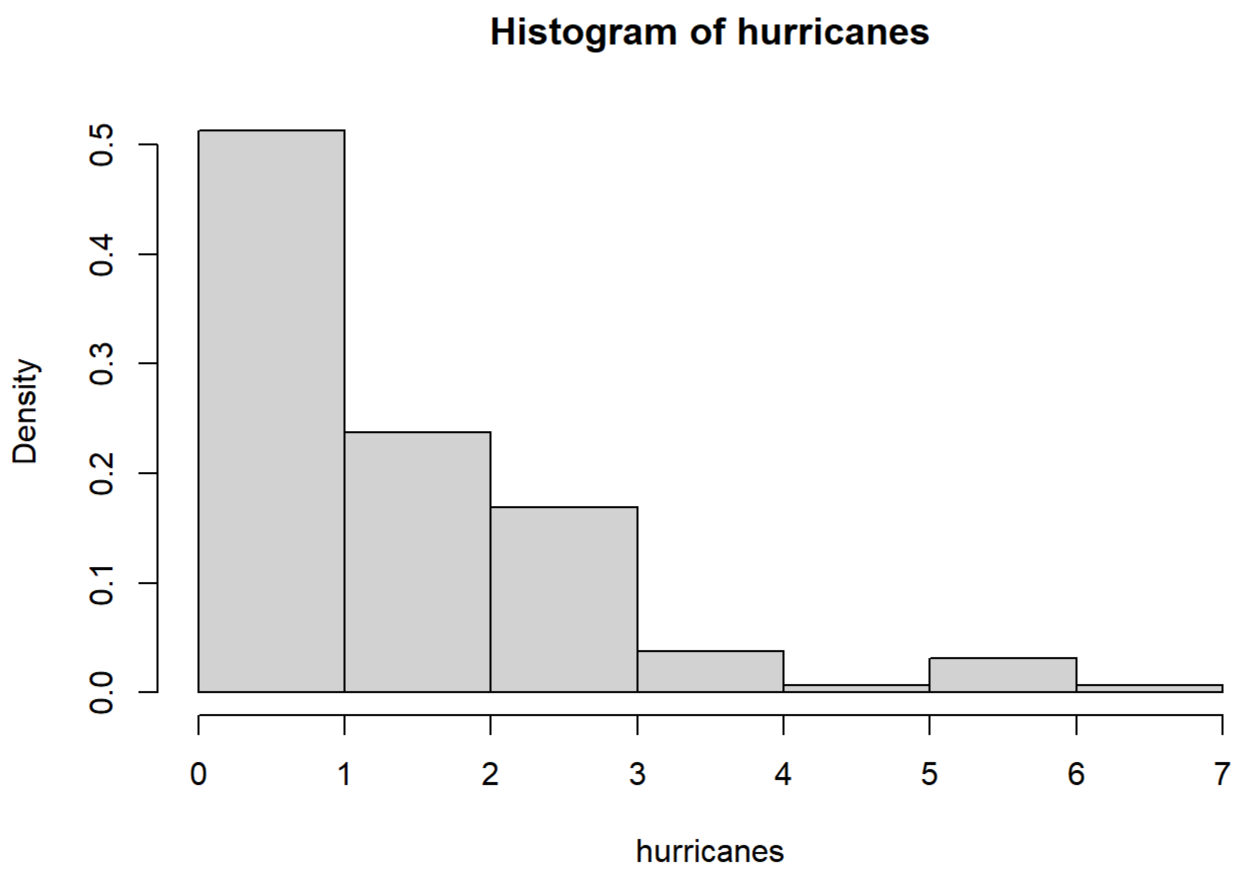
```
[1] 0.029701
```

Tropical Cyclones

```
us <- read.table("US.txt")

colnames(us) <- us[1,]
us <- us[-1,]
```

question 1

```
hurricanes <- as.numeric(us$US)

hist(hurricanes, prob = TRUE)
```




```
# poisson distribution is used for rare events and also
# count data
```

question 2

```
lambda <- mean(hurricanes)
lambda
```

```
[1] 1.69375
```

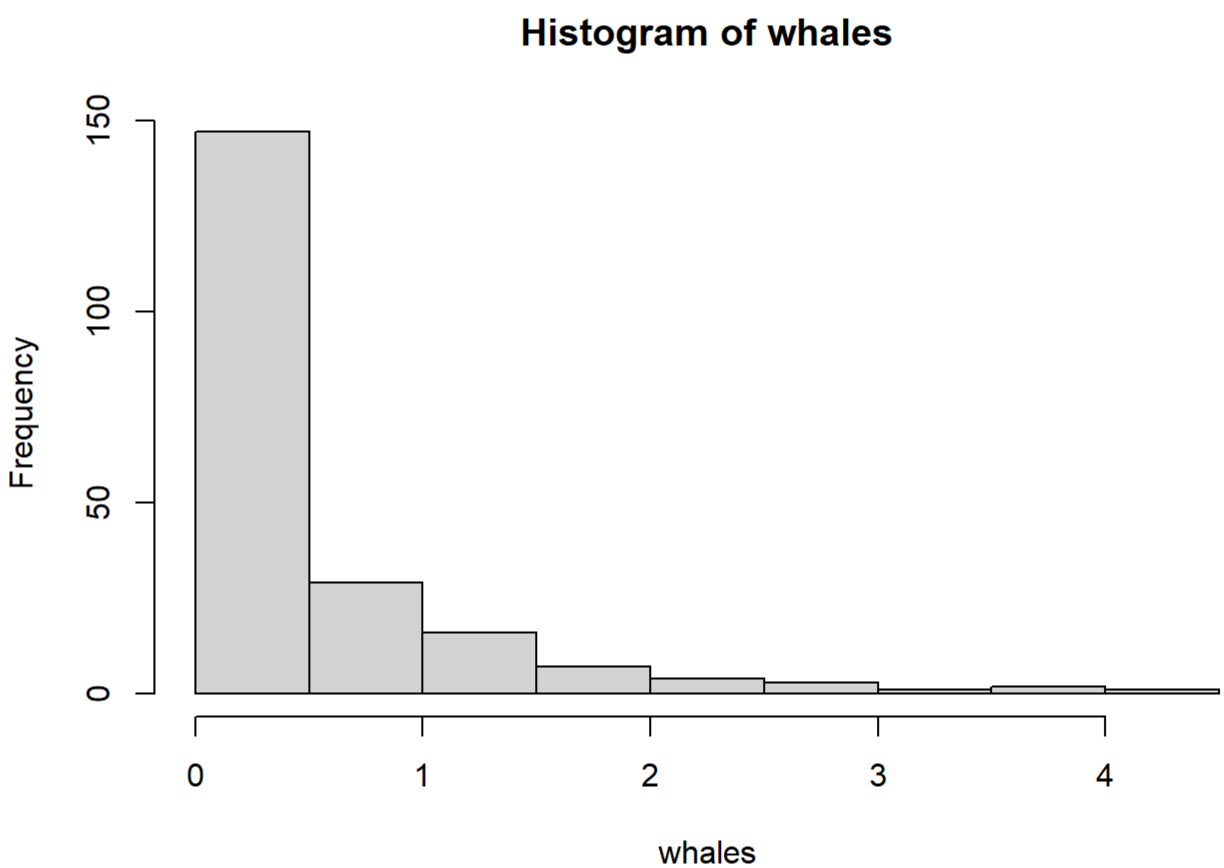
```
## lamba is the mean and also the variance of poisson
## distribution
```

Whales

```
whales <- read.table("whales.txt")

whales <- as.numeric(whales$V1)
```

Question 1

```r
hist(whales)
```



Question 2

Yes a gamma distribution is plausible because swimming can't be negative.

Question 3

```r
mean_bar <- mean(whales)
s2 <- var(whales)

alpha_mom <- mean_bar^2/s2
beta_mom <- mean_bar/s2
alpha_mom
```

```
[1] 0.7953685
```

```r
beta_mom
```

```
[1] 1.312491
```

Question 4

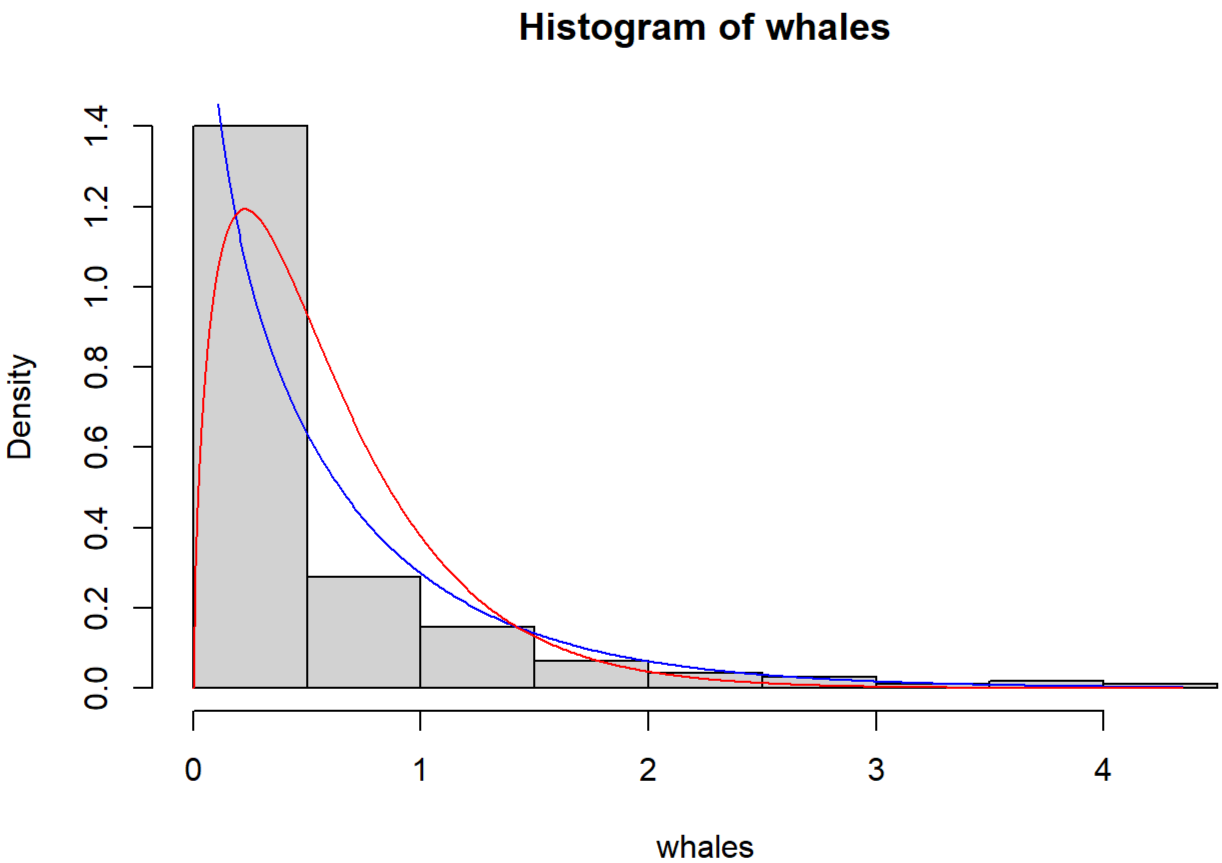
```
## maximum likelihood

fit_gamma_ml <- fitdist(whales, "gamma")
fit_gamma_ml
```

```
Fitting of the distribution ' gamma ' by maximum likelihood
Parameters:
      estimate Std. Error
shape 1.595511  0.1423970
rate  2.632771  0.2754974
```

```
alpha_ml <- 1.595511
beta_ml <- 2.632771

## The values are not the same but different
```

Question 5

```
hist(whales, prob = TRUE)

x <- seq(0, max(whales),length.out = 500)

lines(x, dgamma(x, alpha_mom, rate = beta_mom),
      col = "blue")

lines(x, dgamma(x, alpha_ml, rate = beta_ml),
      col = "red")
```


Histogram of whales
Density
whales
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
0
1
2
3
4